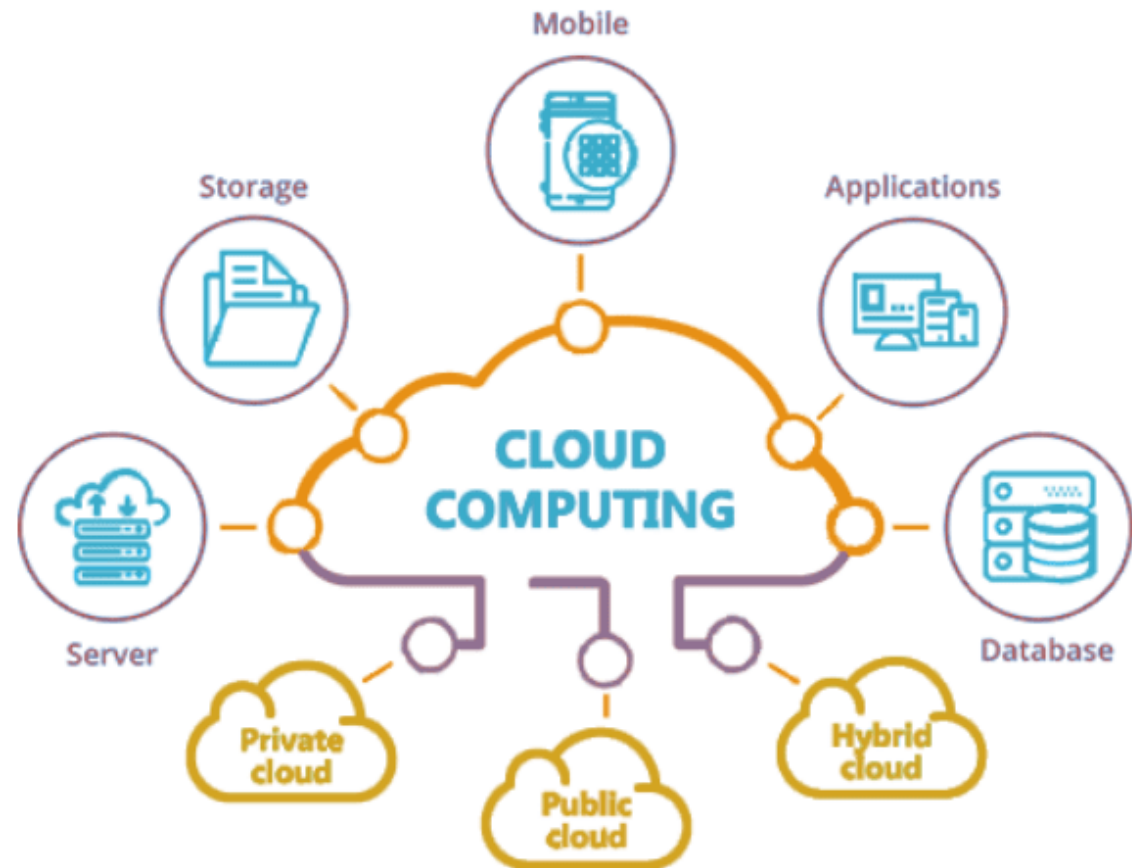


Cloud Computing



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Section 1



Learning Contract & Basic Concepts of Cloud Computing

Outline



- Learning Contract
- Learning Objectives
- Introduction
- Cloud Computing Definition
- Types of Cloud Computing
- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)
- Management of Cloud Services
- Deployment Models of Cloud Computing
- Levels of Abstraction
- Exercise
- Reference

Learning Contract



- Arrive on time
- Dress neatly and politely
- If not present, must permission from the lecturer
- Evaluation:
 - ✓ CPMK 1 = 25% (Task)
 - ✓ CPMK 2 = 20% (Quiz)
 - ✓ CPMK 3 = 25% (Mid Exam)
 - ✓ CPMK 4 = 30% (Final Exam)
- Rating Score:
 - ✓ A=85-100
 - ✓ B=75-84
 - ✓ C=65-74
 - ✓ D=55-64
 - ✓ E<=54
- Certificate Score: 90-100

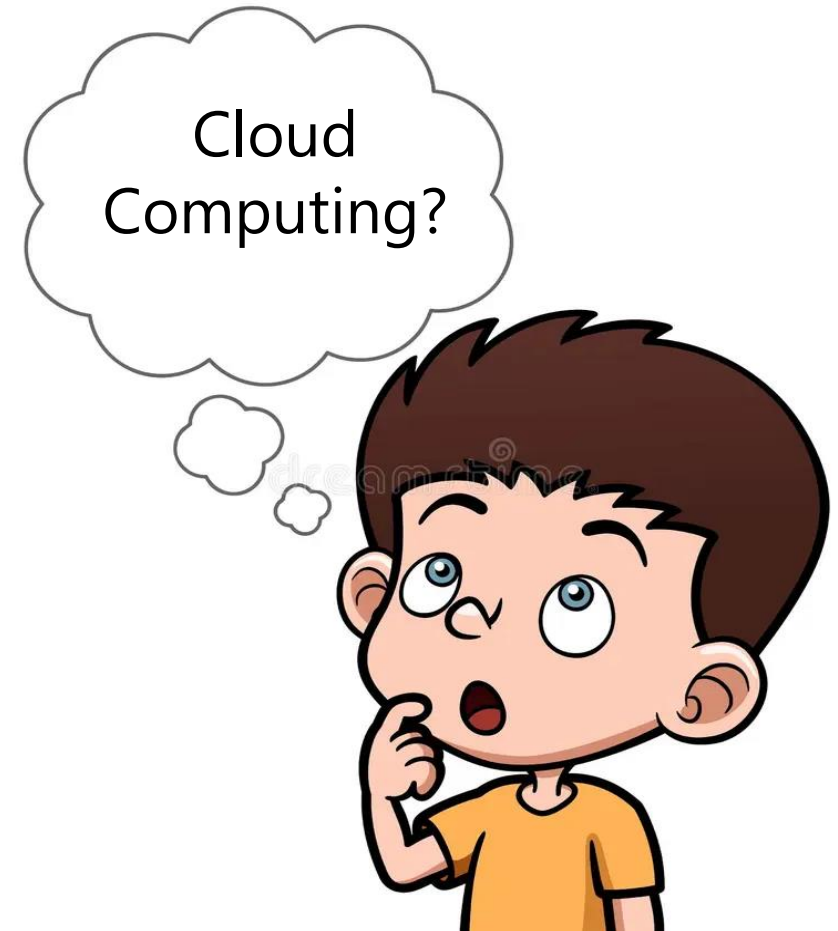
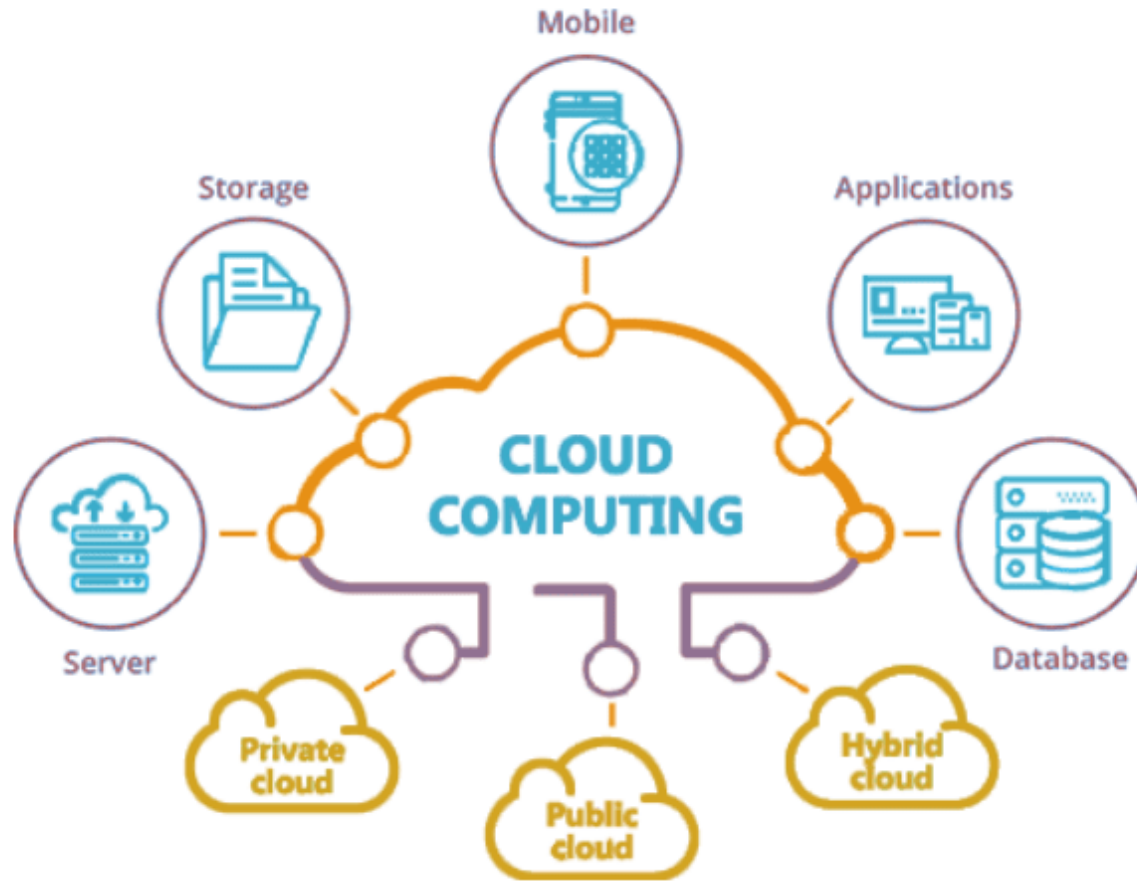


Learning Objectives

After studying this course session, students will be able to:

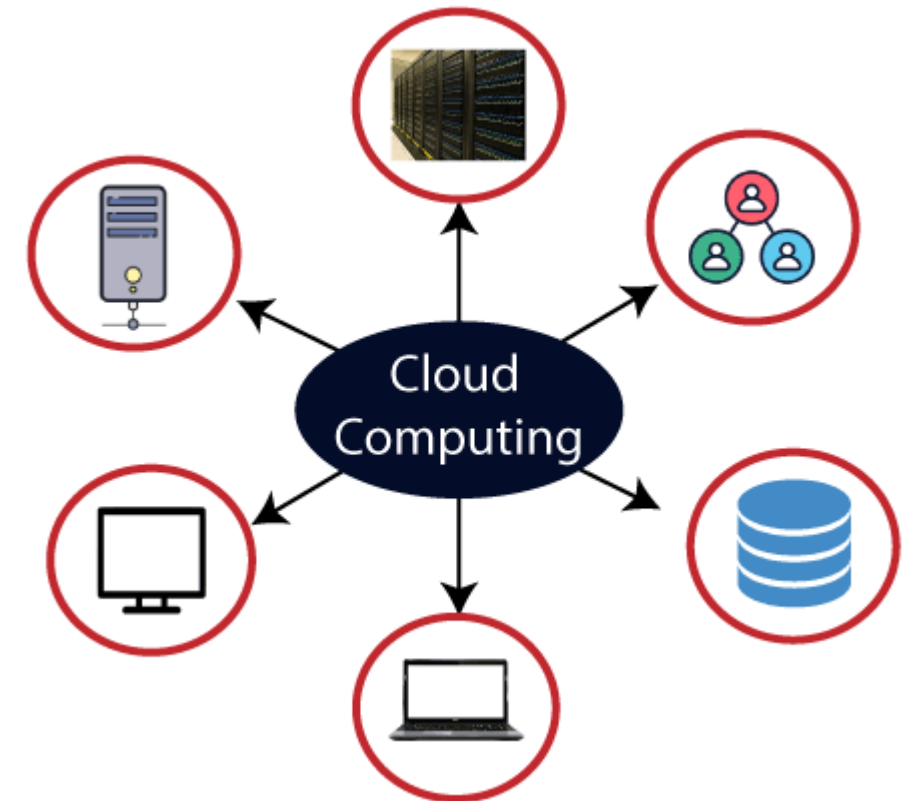
- Understand about basic concept of cloud computing
- Explain about how cloud computing works

Introduction



Cloud Computing Definition

Cloud computing is a model for delivering computing services such as **servers, storage, databases, networking, software, analytics**, and **intelligence** over the **internet** (the "cloud"). This allows for flexible resources, rapid innovation, and economies of scale.





Types of Cloud Computing

Cloud computing is generally categorized into **three main types** based on the **service models** it offers. These types define how the **cloud services** are delivered and managed:

1. **Infrastructure** as a Service (**IaaS**)
2. **Platform** as a Service (**PaaS**)
3. **Software** as a Service (**SaaS**)

Infrastructure as a Service (IaaS)



- **Definition:** IaaS provides the basic building blocks of cloud IT, offering virtualized computing resources such as **virtual machines, storage, networks,** and **load balancers** over the internet.
- **Users:** System administrators, IT managers, and developers who need control over infrastructure but don't want to manage physical hardware.
- **Examples:** Amazon Web Services (AWS EC2), Microsoft Azure Virtual Machines, Google Cloud Compute Engine.

Infrastructure as a Service (IaaS)



- **Use case:**
 - Hosting websites and web applications
 - Backup and disaster recovery solutions
 - Data storage and analytics
 - Development and testing environments

Platform as a Service (PaaS)



- **Definition:** PaaS provides a platform that allows developers to build, deploy, and manage applications without worrying about underlying infrastructure like servers, storage, or networking. It abstracts infrastructure management and provides development tools, database management, and middleware.
- **Users:** Developers and application architects who want to focus on coding and application development without managing infrastructure.
- **Examples:** Google App Engine, Microsoft Azure App Service, Heroku.

Platform as a Service (PaaS)



- **Use case:**
 - Developing web or mobile applications
 - Hosting microservices and APIs
 - Automating development workflows
 - Testing and deployment of applications

Software as a Service (SaaS)



- **Definition:** SaaS delivers fully developed software applications over the internet, where users can access the software via a web browser without installing or maintaining it on their own devices. The service provider manages everything, including the infrastructure, application, and data storage.
- **Users:** End-users (businesses or individuals) who want to use ready-made software without worrying about technical maintenance.
- **Examples:** Google Workspace (Gmail, Docs), Salesforce, Microsoft 365, Dropbox.

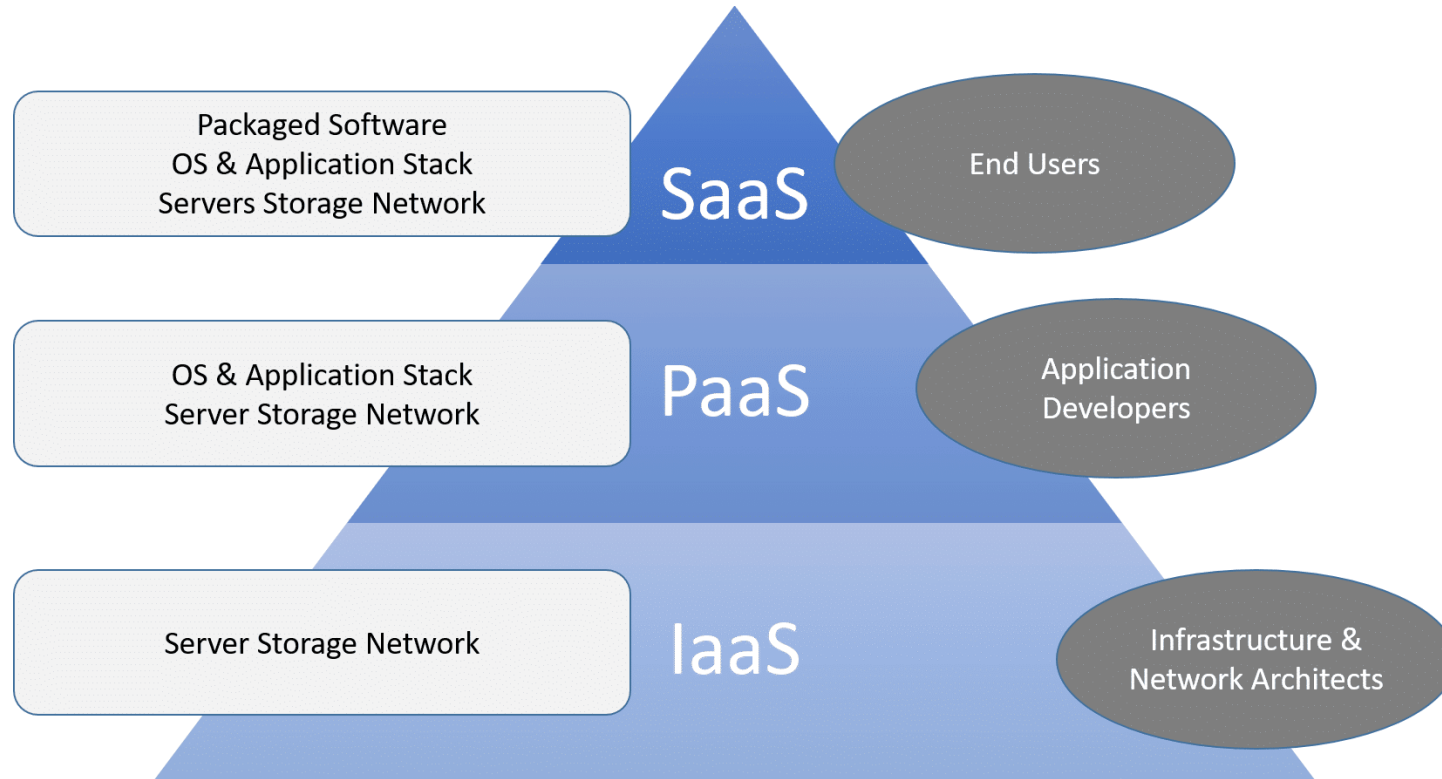
Software as a Service (SaaS)



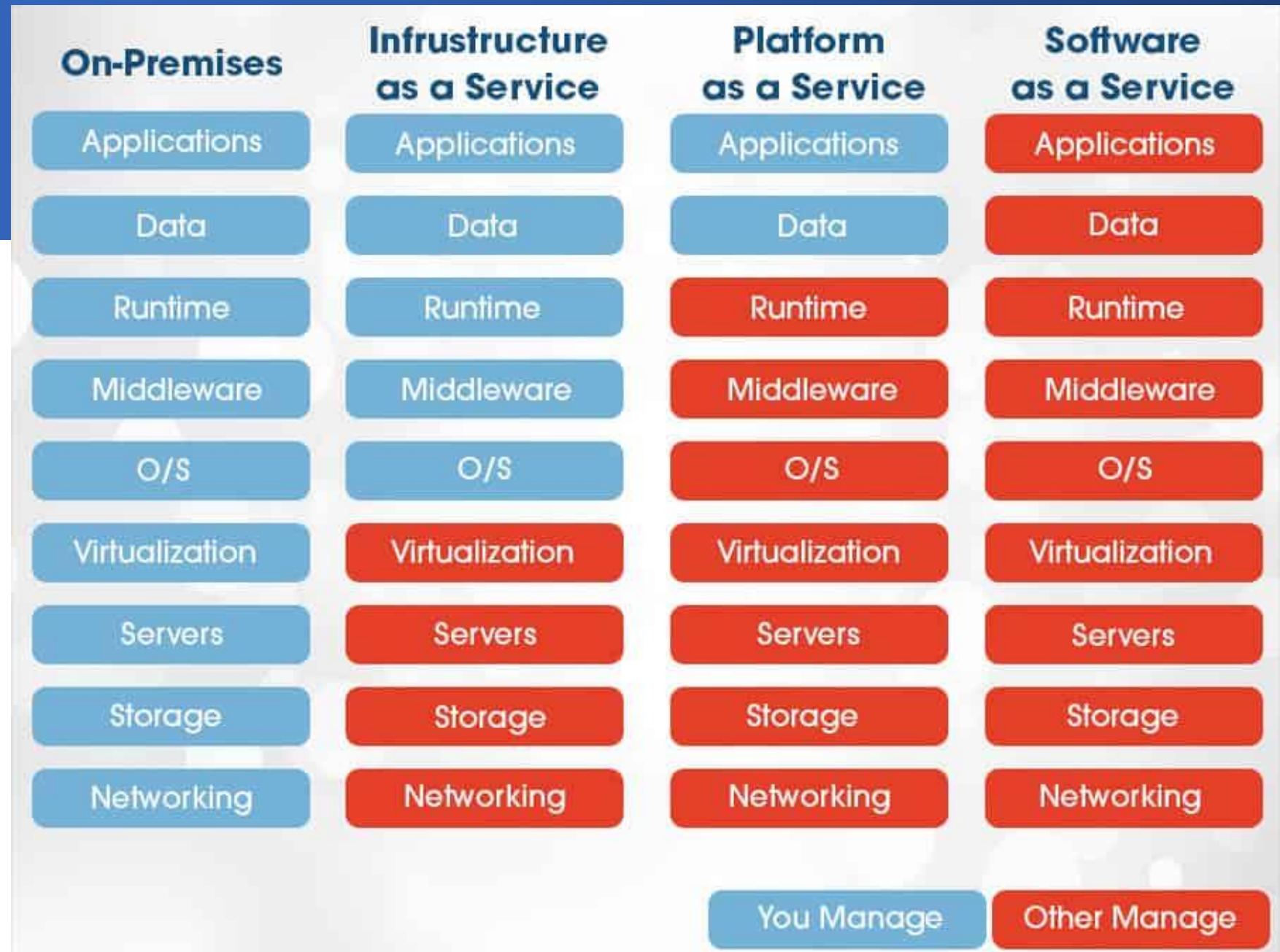
- **Use case:**
 - Email services (e.g., Gmail, Outlook)
 - Customer relationship management (CRM) systems
 - Enterprise resource planning (ERP)
 - Collaboration tools (e.g., Slack, Zoom)

Types of Cloud Computing

Cloud Service Models



Management of Cloud Services



Deployment Models of Cloud Computing



1. Public Cloud:

- Owned and operated by a third-party cloud service provider.
- Shared resources among multiple customers.
- Examples: AWS, Google Cloud, Microsoft Azure.
- Best for: General-purpose, cost-effective solutions with scalability.

Deployment Models of Cloud Computing



2. Private Cloud:

- Cloud infrastructure dedicated to a single organization, either managed in-house or by a third party.
- Higher control over security and customization.
- Best for: Enterprises with strict security or regulatory requirements.

Deployment Models of Cloud Computing



3. Hybrid Cloud:

- Combines public and private clouds, allowing data and applications to be shared between them.
- Provides greater flexibility, optimized costs, and better data management.
- Best for: Workloads that require both public cloud scalability and private cloud security.

Deployment Models of Cloud Computing

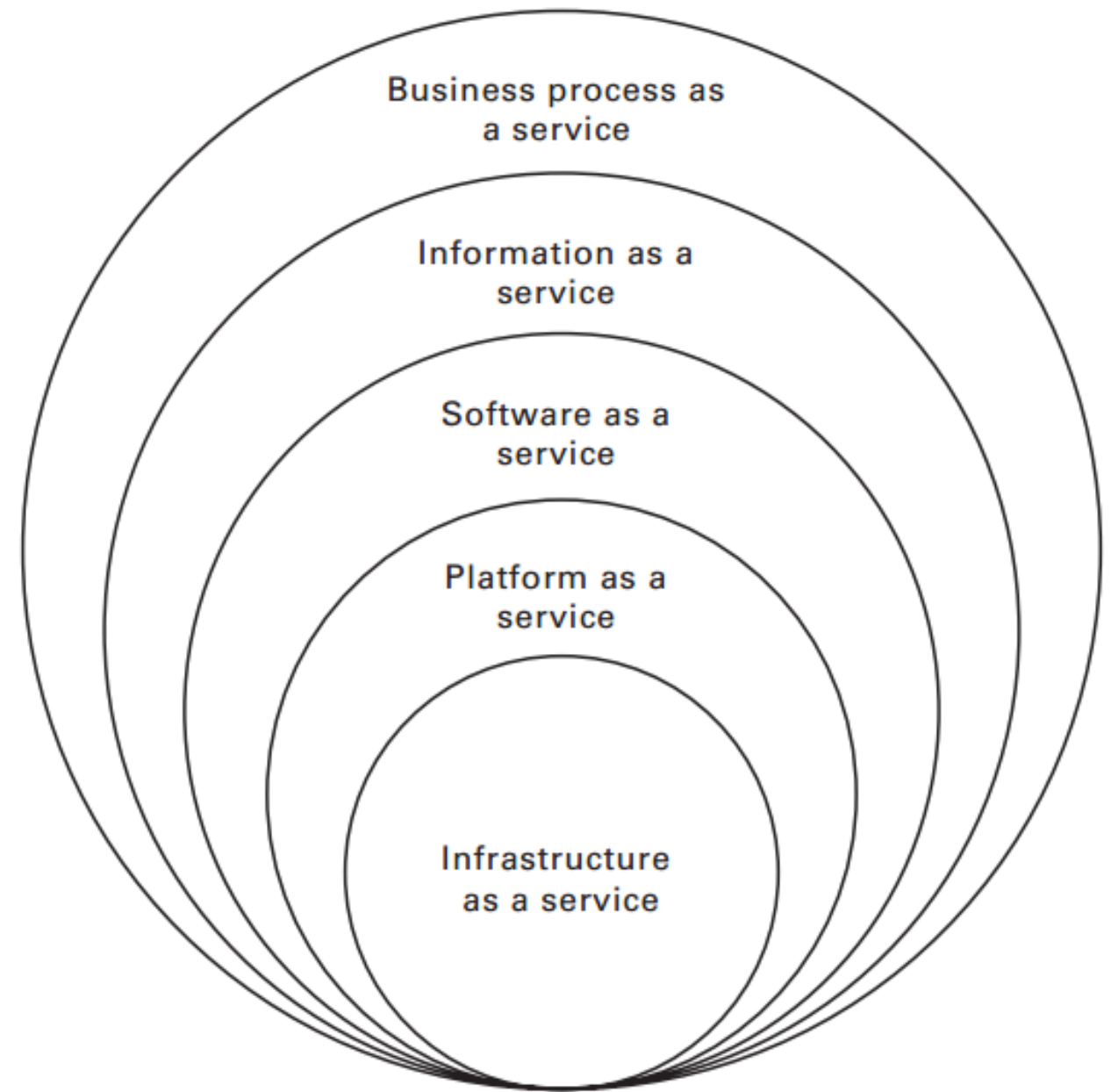


4. Community Cloud:

- A cloud shared by several organizations with common goals or requirements, such as security policies or industry standards.
- Best for: Organizations within a specific industry or government sector.

Levels of Abstraction

- Service models can be viewed in terms of increased levels of abstraction.
- As you move up the stack to the top, you have the highest form of abstraction at the business process as-a-service (BPaaS) level.





Q & A

Reference



- Michael J. Kavis. "Architecting the Cloud: Design Decisions for Cloud Computing Service Models", Wiley
- Thomas Erl, Zaigham Mahmood, Ricardo Puttini, "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl", Prentice Hall



Thank You